

# Drawing Regular Polygons by Side and Angle Measurements

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A **polygon** is a closed plane figure formed by line segments.

# Regular Polygon

A **regular polygon** has all sides equal in length and all interior angles equal in measure.

# Regular Pentagons

A regular polygon with 5 sides is called a **regular pentagon**.

$$n = 5$$

$$\text{Interior angle} = \frac{180(5 - 2)^\circ}{5} = 108^\circ$$

# Regular Hexagons

A regular polygon with 6 sides is called a **regular hexagon**.

$$n = 6$$

$$\text{Interior angle} = \frac{180(6 - 2)^\circ}{6} = 120^\circ$$

# Regular Octagons

A regular polygon with 8 sides is called a **regular octagon**.

$$n = 8$$

$$\text{Interior angle} = \frac{180(8 - 2)^\circ}{8} = 135^\circ$$

# Regular Decagons

A regular polygon with 10 sides is called a **regular decagon**.

$$n = 10$$

$$\text{Interior angle} = \frac{180(10 - 2)}{10} = 144^\circ$$

# Interior Angle Formula

The measure of each interior angle of a regular polygon is

$$\frac{180(n - 2)^{\circ}}{n}$$

where  $n$  is the number of sides.



# Examples

## Example 1

A polygon has 5 equal sides, each measuring 6 cm.

It is a **regular pentagon**.

$$n = 5, \quad s = 6 \text{ cm}$$

## Example 2

A polygon has 6 equal sides, each measuring 4 cm.

It is a **regular hexagon**.

$$n = 6, \quad s = 4 \text{ cm}$$

## Example 3

A polygon has 8 equal sides, each measuring 9 cm.

It is a **regular octagon**.

$$n = 8, \quad s = 9 \text{ cm}$$

# Practice Exercises

# Practice 1

Identify the type and name of the polygon.

A polygon has 5 sides, each measuring 6 cm.

## Practice 2

Identify the type and name of the polygon.

A polygon has 6 sides, each measuring 4 cm.

## Practice 3

Identify the type and name of the polygon.

A polygon has 8 sides, each measuring 9 cm.



# Practice 4

Identify the type and name of the polygon.

A polygon has 10 sides, each measuring 5 cm.

## Practice 5

Identify the type and name of the polygon.

A polygon has 5 sides with measures 4 cm, 4 cm, 4 cm, 4 cm, and 4 cm.

## Practice 6

Identify the type and name of the polygon.

A polygon has 6 sides with measures 7 cm, 7 cm, 7 cm, 7 cm, 7 cm, and 7 cm.

# Practice 7

Identify the type and name of the polygon.

A polygon has 8 sides, each measuring 3 cm.

# Practice 8

Identify the type and name of the polygon.

A polygon has 10 sides, each measuring 8 cm.

## Practice 9

Identify the type and name of the polygon.

A polygon has 5 equal sides, each measuring 10 cm.

## Practice 10

Identify the type and name of the polygon.

A polygon has 6 equal sides, each measuring 12 cm.

# Activities



# Activity 1

Identify the type and name of the polygon.

A polygon has 5 sides, each measuring 8 cm.

## Activity 2

Identify the type and name of the polygon.

A polygon has 6 sides, each measuring 10 cm.

# Activity 3

Identify the type and name of the polygon.

A polygon has 8 sides, each measuring 7 cm.

# Activity 4

Identify the type and name of the polygon.

A polygon has 10 sides, each measuring 3 cm.

# Activity 5

Identify the type and name of the polygon.

A polygon has 5 sides with measures 9 cm, 9 cm, 9 cm, 9 cm, and 9 cm.

# Activity 6

Identify the type and name of the polygon.

A polygon has 6 sides, each measuring 5 cm.

# Activity 7

Identify the type and name of the polygon.

A polygon has 8 sides, each measuring 11 cm.

# Activity 8

Identify the type and name of the polygon.

A polygon has 10 sides, each measuring 6 cm.



## Activity 9

Identify the type and name of the polygon.

A polygon has 5 equal sides, each measuring 14 cm.

# Activity 10

Identify the type and name of the polygon.

A polygon has 6 equal sides, each measuring 9 cm.

Thank you!